

Jacob Eli Turner

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EDUCATION

West Virginia University

Ph.D. in Physics

Advisor: Dr. Maura McLaughlin

Morgantown, WV, USA

Aug 2018 – Aug 2023 (Defended)/Dec 2023 (Conferred)

Oberlin College

B.A. with Honors in Physics

Advisor: Dr. Dan Stinebring

Oberlin, OH, USA

Aug 2013 – May 2017

PROFESSIONAL EMPLOYMENT AND RESEARCH EXPERIENCE

Cornell University/National Radio Astronomy Observatory

Jansky Postdoctoral Fellow

Ithaca, NY, USA

August 2026–Present

Green Bank Observatory

Postdoctoral Fellow

Green Bank, WV, USA

August 2023–August 2026

Used cyclic spectroscopy to study the small-scale structure of the Milky Way via pulsar scintillation. Assisted in the development and testing of the world's first (and currently only) cyclic spectroscopy telescope backend. Trained Green Bank Telescope observers and reviewed technical justifications for observing proposals. Served as the on-call scientist for observations. Organized colloquia and lunch talks. Supervised by Ryan Lynch.

West Virginia University, Department of Physics & Astronomy

Graduate Research Assistant

Graduate Teaching Assistant

Visiting Scholar

Morgantown, WV, USA

2019–2023

2018–2019

2018

University of Wisconsin-Milwaukee, Department of Physics

Research Analyst

Milwaukee, WI, USA

2017

California Institute of Technology, Department of Astronomy

Summer Research Intern, Visiting Undergraduate Research Program

Pasadena, CA, USA

2016

Oberlin College, Department of Physics & Astronomy

Drop-In Tutor

Undergraduate Research Assistant

Undergraduate Teaching Assistant

Oberlin, OH, USA

2017

2015–2017

2015

PUBLICATIONS (37 TOTAL, 8 LEAD-AUTHOR)

[NASA SciX Page](#)

NOTE: Authors with asterisks indicate students ranging from high school to graduate school working under my supervision

Lead-Author Publications

8. [Constraining Scattering Medium Geometry with Cyclic Spectroscopy](#)

Turner, J. E., 2026, ApJL, 1008, L35

7. [Evaluating the Prospects of Cyclic Deconvolution across 312 Pulsars](#)

Turner, J. E., 2026, ApJ, 1001, 231

6. [Pulsar Cyclic Spectroscopy in the Partial-Deconvolution Regime: Benefits & Limitations](#)

Turner, J. E., Dolch, T., Demorest, P. B., Lynch R. S., Stinebring, D. R., Jessup C., Jones, N., and Scheithauer, C., 2025, ApJ, 989, 228

5. [The Pulsar Science Collaboratory: Multi-Epoch Scintillation Studies of Pulsars](#)
Turner, J. E., Lebron Medina, J. G. *, Zelensky, Z. *, Gustavson, K. A., Marx, J., Kothapalli, M. *, Cruz Vega, L. D. *, Lee, A. *, Figueroa, C. B. *, Reichart, D. E., Haislip, J. B., Kouprianov, V. V., White, S., Ghigo, F., Heatherly, S. A., and McLaughlin, M. A., 2024, ApJ, 977, 205
4. [A Cyclic Spectroscopy Scintillation Study of PSR B1937+21 I. Demonstration of Improved Scintillometry](#)
Turner, J. E., Dolch, T., Cordes, J. M., Ocker, S. K., Stinebring, D. R., Chatterjee, S., McLaughlin, M. A., Catlett, V. E., Jessup C., Jones, N., and Scheithauer, C., 2024, ApJ, 972, 16
3. [A Simultaneous Dual-Frequency Scintillation Arc Survey of Six Bright Canonical Pulsars Using the Upgraded Giant Metrewave Radio Telescope](#)
Turner, J. E., Joshi, B.C., McLaughlin, M. A., and Stinebring, D. R., 2024, ApJ, 961, 101
2. [Scattering Delay Mitigation in High Accuracy Pulsar Timing: Cyclic Spectroscopy Techniques](#)
Turner, J. E., Stinebring, D. R., McLaughlin, M. A., Archibald, A. M., Dolch, T., and Lynch, R. S., 2023, ApJ, 944, 191
1. [The NANOGrav 12.5-Year Data Set: Monitoring Interstellar Scattering Delays](#)
Turner, J. E., et al. (36 authors), 2021, ApJ, 917, 10

Student-Led Publications

1. [An Extreme Scattering Event Toward PSR J2313+4253](#)
Zelensky, Z. *, Turner, J. E., Lebron Medina, J. G. *, Reichart, D. E., Haislip, J. B., Kouprianov, V. V., White, S., Ghigo, F., Heatherly, S. A., and McLaughlin, M. A., 2025, Submitted to ApJ

Other Publications

28. [The NANOGrav 15 yr Dataset: Targeted Searches for Supermassive Black Hole Binaries](#)
Agarwal, N. et al., (122 authors, including Turner, J. E.), 2026, ApJ, 998, 1
27. [Inferring \$M_{\text{BH}}\text{-}M_{\text{bulge}}\$ Evolution from the Gravitational-wave Background](#)
Cayenne, M. et al., (107 authors, including Turner, J. E.), 2026, ApJ, 997, 2
26. [The NANOGrav 15 yr Data Set: Search for Gravitational-wave Memory](#)
Agazie, G. et al., (105 authors, including Turner, J. E.), 2025, ApJ, 987, 1
25. [The NANOGrav 15 yr Data Set: Harmonic Analysis of the Pulsar Angular Correlations](#)
Agazie, G. et al., (107 authors, including Turner, J. E.), 2025, ApJ, 985, 1
24. [The NANOGrav 15 yr dataset: Posterior predictive checks for gravitational-wave detection with pulsar timing arrays](#)
Agazie, G. et al., (104 authors, including Turner, J. E.), 2025, PhRvD, 111, 4
23. [The NANOGrav 15 yr Data Set: Running of the Spectral Index](#)
Agazie, G. et al., (105 authors, including Turner, J. E.), 2025, ApJL, 978, 2
22. [The NANOGrav 15 Yr Data Set: Removing Pulsars One by One from the Pulsar Timing Array](#)
Agazie, G. et al., (105 authors, including Turner, J. E.), 2025, ApJ, 978, 2
21. [The NANOGrav 15 yr Data Set: Looking for Signs of Discreteness in the Gravitational-wave Background](#)
Agazie, G. et al., (100 authors, including Turner, J. E.), 2025, ApJ, 978, 1
20. [Scintillation Bandwidth Measurements from 23 Pulsars from the AO327 Survey](#)
Sheikh, S., Brown, G. C., MacTaggart, J., Nguyen, T., Fletcher, W. D., Jones, B. L., Koller, E., Petrus, V., Pighini, K. F., Rosario, G., Smedile, V. A., Stone, A. T., You, S., McLaughlin, M. A., Turner, J. E., Deneva, J. S., Lam, M. T., and Shapiro-Albert, B. J., 2024, ApJ, 976, 2
19. [NANOGrav 15-year gravitational-wave background methods](#)
Johnson, A. D. et al., (98 authors, including Turner, J. E.), 2024, PhRvD, 109, 10

18. [Comparing Recent Pulsar Timing Array Results on the Nanohertz Stochastic Gravitational-wave Background](#)
The International Pulsar Timing Array Collaboration, et al., (244 authors, including **Turner, J. E.**), 2024, ApJ, 966, 1
17. [The NANOGrav 12.5 yr Data Set: A Computationally Efficient Eccentric Binary Search Pipeline and Constraints on an Eccentric Supermassive Binary Candidate in 3C 66B](#)
Agazie, G., et al., (89 authors, including **Turner, J. E.**), 2024, ApJ, 963, 2
16. [The NANOGrav 12.5 yr Data Set: Search for Gravitational Wave Memory](#)
Agazie, G., et al., (91 authors, including **Turner, J. E.**), 2024, ApJ, 963, 1
15. [How to Detect an Astrophysical Nanohertz Gravitational Wave Background](#)
Bécsy, B., et al., (96 authors, including **Turner, J. E.**), 2023, ApJ, 959, 1
14. [The NANOGrav 15 yr Data Set: Search for Anisotropy in the Gravitational-wave Background](#)
Agazie, G., et al., (93 authors, including **Turner, J. E.**), 2023, ApJL, 956, 1
13. [The NANOGrav 15 yr Data Set: Constraints on Supermassive Black Hole Binaries from the Gravitational-wave Background](#)
Agazie, G., et al., (99 authors, including **Turner, J. E.**), 2023, ApJL, 952, 2
12. [The NANOGrav 15 yr Data Set: Bayesian Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries](#)
Agazie, G., et al., (99 authors, including **Turner, J. E.**), 2023, ApJL, 951, 2
11. [The NANOGrav 15 yr Data Set: Search for Signals from New Physics](#)
Afzal, A., et al., (124 authors, including **Turner, J. E.**), 2023, ApJL, 951, 1
10. [The NANOGrav 15 yr Data Set: Detector Characterization and Noise Budget](#)
Agazie, G., et al., (92 authors, including **Turner, J. E.**), 2023, ApJL, 951, 1
9. [The NANOGrav 15 yr Data Set: Observations and Timing of 68 Millisecond Pulsars](#)
Agazie, G., et al., (101 authors, including **Turner, J. E.**), 2023, ApJL, 951, 1
8. [The NANOGrav 15 yr Data Set: Evidence for a Gravitational-wave Background](#)
Agazie, G., et al., (115 authors, including **Turner, J. E.**), 2023, ApJL, 951, 1
7. [Searching for continuous Gravitational Waves in the second data release of the International Pulsar Timing Array](#)
Falxa, M., et al., (127 authors, including **Turner, J.**), 2023, MNRAS, 521, 4
6. [Searching For Gravitational Waves From Cosmological Phase Transitions with the NANOGrav 12.5-year Dataset](#)
Arzoumanian, Z., et al., (64 authors, including **Turner, J. E.**), 2021, PRL, 127, 251302
5. [The NANOGrav 12.5-year data set: Search for Non-Einsteinian Polarization Modes in the Gravitational-Wave Background](#)
Arzoumanian, Z., et al., (71 authors, including **Turner, J. E.**), 2021, ApJL, 923, L22
4. [The NANOGrav 12.5-year Data Set: Search For An Isotropic Stochastic Gravitational-Wave Background](#)
Arzoumanian, Z., et al. (61 authors, including **Turner, J. E.**), 2020, ApJ, 905, L34
3. [The NANOGrav 11-Year Data Set: Evolution of Gravitational Wave Background Statistics](#)
Hazboun, J. S., et al. (63 authors, including **Turner, J. E.**), 2020, ApJ, 890, 108
2. [The NANOGrav 11 yr Data Set: Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries](#)
Aggarwal, K., et al. (63 authors, including **Turner, J. E.**), 2019, ApJ, 880, 116
1. [A Second Chromatic Timing Event of Interstellar Origin toward PSR J1713+0747](#)
Lam, M. T., Ellis, J. A., Grillo, G., Jones, M. L., Hazboun, J. S., Brook, P. R., **Turner, J. E.**, et al. (37 authors), 2018, ApJ, 861, 132

INVITED TALKS

National Radio Astronomy Observatory <i>Pulsar Cyclic Spectroscopy as a Probe of the Interstellar Medium</i>	Socorro, NM, USA October 2026
University of New Mexico <i>Pulsar Cyclic Spectroscopy as a Probe of the Interstellar Medium</i>	Albuquerque, NM, USA October 2026
Cornell University <i>Pulsar Cyclic Spectroscopy as a Probe of the Interstellar Medium</i>	Ithaca, NY, USA September 2026
Green Bank Observatory Community Zoom <i>Constraining Scattering Medium Geometry with Cyclic Spectroscopy</i>	Green Bank, WV, USA June 2026
Scintillometry Workshop 2025 (McGill University) <i>Exploring The Benefits and Feasibility of Cyclic Spectroscopy in Different Deconvolution Regimes</i>	Montreal, Quebec, Canada October 2025
Oregon State University <i>Pulsar Cyclic Spectroscopy as a Probe of the Interstellar Medium & Gravitational Waves</i>	Corvallis, OR, USA August 2025
GRASP Lecture Series (Remote) <i>Pulsar Cyclic Spectroscopy as a Probe of the Interstellar Medium & Gravitational Waves</i>	Cape Town, South Africa August 2025
Green Bank Observatory Community Zoom <i>An Extreme Scattering Event Towards PSR B2310+42</i>	Green Bank, WV, USA June 2025
Georgia State University <i>Pulsar Cyclic Spectroscopy as a Probe of the Interstellar Medium & Gravitational Waves</i>	Atlanta, GA, USA April 2025
University of Kansas (Remote) <i>Pulsar Cyclic Spectroscopy as a Probe of the Interstellar Medium & Gravitational Waves</i>	Lawrence, KS, USA April 2025
Scintillometry Workshop 2024 (Florida Space Institute) <i>The Green Bank Observatory Real-Time Cyclic Spectroscopy Backend</i>	Orlando, FL, USA October 2024
Florida Space Institute <i>Using Cyclic Spectroscopy to Study the Interstellar Medium with Pulsar Timing Arrays</i>	Orlando, FL, USA September 2024
Florida Institute of Technology <i>Using Cyclic Spectroscopy to Study the Interstellar Medium with Pulsar Timing Arrays</i>	Orlando, FL, USA September 2024
University of Dallas <i>Two Paths to Radio Astronomy</i>	Dallas, TX, USA April 2024
McDaniel College <i>Characterizing the Interstellar Medium through Radio Observations of Pulsars</i>	Westminster, MD, USA November 2023
Green Bank Observatory (Remote) <i>Correcting for Interstellar Scattering Delays in Millisecond Pulsars</i>	Green Bank, WV, USA November 2020
Oberlin College <i>Detecting Gravitational Waves with Pulsars: Removing the Effects of the Interstellar Medium</i>	Oberlin, OH, USA April 2017

CONTRIBUTED CONFERENCE TALKS

National Radio Astronomy Observatory/Green Bank Observatory Postdoc Symposium <i>Constraining Scattering Medium Geometry and Turbulence with Cyclic Spectroscopy</i>	NRAO April 2026
Green Bank Observatory Internal Symposium <i>Constraining Scattering Medium Geometry and Turbulence with Cyclic Spectroscopy</i>	Green Bank Observatory April 2026
North American Nanohertz Observatory for Gravitational Waves (NANOGrav) Conference <i>Evaluating the Prospects of Cyclic Deconvolution in Millisecond Pulsars</i>	University of Montana November 2025
Pulsar 2025 Conference <i>Pulsar Cyclic Spectroscopy as a Probe of the Interstellar Medium & Gravitational Waves</i>	Sardinia, Italy September 2025
International Pulsar Timing Array Conference <i>Pulsar Cyclic Spectroscopy in the Partial-Deconvolution Regime: Benefits & Limitations</i>	Caltech June 2025
National Radio Astronomy Observatory/Green Bank Observatory Postdoc Symposium 2024 <i>An Extreme Scattering Event Towards PSR B2310+42</i>	NRAO May 2025
Green Bank Observatory Internal Symposium <i>An Extreme Scattering Event Towards PSR B2310+42</i>	Green Bank Observatory May 2025
North American Nanohertz Observatory for Gravitational Waves (NANOGrav) Conference <i>The Pulsar Science Collaboratory: Multi-Epoch Scintillation Studies</i>	University of Michigan October 2024
International Pulsar Timing Array Conference <i>Cyclic Spectroscopy Studies of the ISM in PTA Observing Setups</i>	Sexten Center of Astrophysics June 2024
Fields, Flows, and Filaments in the Magnetic ISM Workshop <i>Cyclic Spectroscopy Studies of the ISM in PTA Observing Setups</i>	Stanford University May 2024
National Radio Astronomy Observatory/Green Bank Observatory Internal Symposium <i>The Pulsar Science Collaboratory: Multi-Epoch Scintillation Studies of Pulsars</i>	Green Bank Observatory May 2024
National Radio Astronomy Observatory/Green Bank Observatory Internal Symposium <i>The Pulsar Science Collaboratory: Multi-Epoch Scintillation Studies of Pulsars</i>	Green Bank Observatory May 2024
National Radio Astronomy Observatory/Green Bank Observatory Postdoc Symposium <i>Using Cyclic Spectroscopy in High-Accuracy Pulsar Timing Efforts</i>	Green Bank Observatory March 2024
Scintillometry Workshop 2023 <i>Using Cyclic Spectroscopy in High-Accuracy Pulsar Timing Efforts</i>	ASIAA November 2023
North American Nanohertz Observatory for Gravitational Waves (NANOGrav) Conference <i>Scattering Delay Mitigation in High Accuracy Pulsar Timing: Cyclic Spectroscopy Techniques</i>	Oregon State University March 2023
241 st American Astronomical Society Meeting <i>Characterizing and Mitigating Scattering Delays in Radio Observations of Pulsars</i>	Seattle, WA, USA January 2023
North American Nanohertz Observatory for Gravitational Waves (NANOGrav) Conference <i>The NANOGrav 12.5-Year Data Set: Monitoring Interstellar Scattering Delays</i>	Cornell University October 2019
International Pulsar Timing Array Conference <i>The NANOGrav 12.5-Year Data Set: Monitoring Interstellar Scattering Delays</i>	NCRA-TIFR June 2019

CONFERENCE POSTERS

247 th American Astronomical Society Meeting <i>Exploring The Benefits and Feasibility of Cyclic Spectroscopy in Different Deconvolution Regimes</i>	Phoenix, AZ, USA January 2026
Scintillometry Workshop 2024 <i>Evidence of an Extreme Scattering Event towards PSR J2313+4253</i>	Florida Space Institute October 2024
243 rd American Astronomical Society Meeting <i>Cyclic Spectroscopy-Aided Studies of the ISM in PTA Observing Setups</i>	New Orleans, LA, USA January 2024
North American Nanohertz Observatory for Gravitational Waves (NANOGrav) Conference <i>Cyclic Spectroscopy-Aided Studies of the ISM in PTA Observing Setups</i>	UBC October 2023
International Pulsar Timing Array Conference <i>The NANOGrav 12.5-Year Data Set: Monitoring Interstellar Scattering Delays</i>	Albuquerque, NM, USA June 2018
NANOGrav Physics Frontiers Center Reverse Site Visit <i>Preliminary Continuous Wave Limits from NANOGrav 11-Year Dataset</i>	West Virginia University October 2017
NANOGrav Physics Frontiers Center Reverse Site Visit <i>NANOGrav Timing Pipeline: Adding a Scattering Delay Correction</i>	West Virginia University October 2017

TEACHING EXPERIENCE

Green Bank Observatory Observing Mentor <i>Green Bank Telescope 25B Observer Training</i>	Green Bank, WV, USA November 2025
Green Bank Observatory Lecturer/Research Mentor <i>Pulsar Science Collaboratory Camp</i>	Green Bank, WV, USA August 2025
Green Bank Observatory Lecturer/Observing Mentor <i>Green Bank Telescope Semester 24B Observer Training</i>	Green Bank, WV, USA October 2024
Green Bank Observatory Lecturer/Research Mentor <i>Green Bank Observatory Single Dish Summer School</i>	Green Bank, WV, USA July 2024
Green Bank Observatory Lecturer/Research Mentor <i>Pulsar Science Collaboratory Camp</i>	Green Bank, WV, USA June 2024
Green Bank Observatory Lecturer/Observing Mentor <i>Green Bank Telescope Semester 24A Observer Training</i>	Green Bank, WV, USA February 2024
West Virginia University Guest Lecturer <i>ASTR 700: Radio Astronomy</i>	Morgantown, WV, USA Spring 2020
West Virginia University Graduate Teaching Assistant <i>PHYS 102L: Introductory Physics 2 Laboratory</i>	Morgantown, WV, USA Spring 2019
West Virginia University Graduate Teaching Assistant <i>PHYS 101L: Introductory Physics 1 Laboratory</i>	Morgantown, WV, USA Fall 2018

Oberlin College
Drop-in Tutor
PHYS 068: Energy Science & Technology

Oberlin, OH, USA
Spring 2017

Oberlin College
Undergraduate Teaching Assistant
PHYS 104: Elementary Physics II Laboratory

Oberlin, OH, USA
Spring 2015

STUDENT RESEARCH MENTORSHIP SUPERVISION

Pulsar Science Collaboratory Research Team Leader, Scintillation Measurement Project 2021–Present

— Students: Juan G. Lebron Medina (Graduate Student, University of Puerto Rico-Mayaguez), Zachary Zelensky (Graduate Student, Texas Tech), Manvith Kothapalli (Undergrad, University of Washington-Seattle), Luis D. Cruz Vega (Undergrad, University of Puerto Rico-Mayaguez), Alexander Lee (Undergrad, University of Washington-Seattle), Caryelis B. Figueroa (Graduate Student, University of Puerto Rico-Mayaguez), Martina Salichs-Maidana (Undergrad, University of Puerto Rico-Mayaguez), Sanjit Subramaniam (High School Student), Katelyn Bryant (Graduate Student, West Virginia University), Dhruva Kalyani (Undergrad, University of Wisconsin-Madison), Adrian Hsu (High School Student), Lahari Ganti (High School Student), Kaito Hasebe (Undergrad, University of Washington-Bothell)

— Authored Peer-Reviewed Paper With 6 Students

— Authored Successful Green Bank Telescope Observing Proposal With 5 Students (Awarded 50 Hours)

— Authored Successful Green Bank Telescope Observing Proposal With 1 Student (Awarded 40 Hours)

Green Bank Observatory REU Summer Student Mentor 2024-2026

— Students: Rachel King (West Virginia University), Dhruva Kalyani (University of Wisconsin-Madison), Lawrence Wise (University of Virginia), Erica Egleton (University of Chicago)

Undergraduate Senior Thesis Project Co-Mentor 2024–2025

— Katelyn Bryant (Undergrad, University of Arkansas)

OUTREACH

Skype A Scientist (over 20 talks given to various elementary, middle, and high schools) 2020–Present

Adopt-A-Physicist 2023–Present

Scientist Presenter for SETI tours at Green Bank Observatory 2024–2026

Pocahontas County Science Fair Judge February 2024

OUTREACH TALKS

Cornell Astronomical Society September 2026
The Search for Extraterrestrial Intelligence: History & Implications

Northern Virginia Astronomy Club May 2026
Neutron Stars: Nature's Most Versatile Laboratories

University of Wisconsin-Madison Astronomy Club February 2026
Neutron Stars: Nature's Most Versatile Laboratories

Rose City Astronomers August 2025
Neutron Stars: Nature's Most Versatile Laboratories

Astronomy on Tap Corvallis August 2025
Neutron Stars: Nature's Most Versatile Laboratories

Green Bank Observatory PING (Physicists Inspiring the Next Generation) Camp <i>Neutron Stars: Nature's Most Versatile Laboratories</i>	July 2025
West Virginia Governor's STEM Institute <i>Neutron Stars: Nature's Most Versatile Laboratories</i>	July 2025
Green Bank Observatory PING (Physicists Inspiring the Next Generation) Camp <i>Using Pulsars to Explore the Universe</i>	July 2024
Pulsar Science Collaboratory (PSC) Talk Series <i>Using Pulsars to Study the Interstellar Medium</i>	April 2024

TELESCOPE TIME ALLOCATIONS

Green Bank Telescope <i>Breaking the Gabor Uncertainty Relation in Slow Pulsars With Cyclic Spectroscopy</i>	GBT26B-267, 10 hours Observation PI
Green Bank Telescope <i>A Scintillation Arc Survey of Seven Scattered Pulsars Using Cyclic Spectroscopy</i>	GBT26B-265, 35 hours Observation PI
Green Bank Telescope <i>Cyclic Spectroscopy of Scattered NANOGrav Pulsars: Pilot for CS Observations</i>	GBT25B-264, 45 hours Observation PI
Green Bank Telescope <i>Tracking A Multiple Order-of-Magnitude Change in Scintillation Towards A Pulsar</i>	GBT25B-040, 40 hours Observation PI
Green Bank Telescope <i>Multi-Hour Scintillation Studies by the PSC</i>	GBT24B-040, 50 hours Observation PI
Green Bank Telescope <i>Cyclic Spectroscopy of Three Pulsars with Considerable Pulse Broadening</i>	GBT24B-039, 45 hours Observation PI
Green Bank Telescope <i>Constraining the Scintillation Constant C_1 in a Scatter-Broadened Pulsar</i>	GBT24A-475, 5 hours Observation PI
Upgraded Giant Metrewave Radio Telescope <i>Examining the Relation Between Scintillation Arc Curvature and Asymmetry</i>	44_035, 25 hours Observation PI
Upgraded Giant Metrewave Radio Telescope <i>Scintillation Arcs and Dispersion Measure Changes: A Follow-up to Pilot Observations</i>	40_019, 24 hours Observation PI
Green Bank Telescope <i>A Cyclic Spectroscopy Pilot Program: Baseband Observations of Three MSPs</i>	GBT20A-588, 12 hours
Upgraded Giant Metrewave Radio Telescope <i>Scintillation Arcs and Dispersion Measure Changes: A Pilot Project</i>	38_041, 24 hours Observation PI

PROFESSIONAL COMMUNITY SERVICE/LEADERSHIP

LOC <i>Scintillometry Workshop 2026</i>	2026
Journal Referee <i>Monthly Notices of the Royal Astronomical Society</i>	2026—Present

Moderator <i>Advancing Pulsar Science in the Era of Modern Interferometers</i>	2026
Journal Referee <i>The Astrophysical Journal</i>	2025–Present
Observing Proposal Scientific Reviewer <i>Upgraded Giant Metrewave Radio Telescope</i>	2024–Present
LOC <i>National Radio Astronomy Observatory/Green Bank Observatory Postdoc Symposium</i>	2024
Colloquium/Science Lunch Talk Organizer <i>Green Bank Observatory Postdoc Symposium</i>	2023–Present
Moderator <i>North American Nanohertz Observatory for Gravitational Waves (NANOGrav) Conference</i>	2022
Moderator <i>North American Nanohertz Observatory for Gravitational Waves (NANOGrav) Conference</i>	2021

MEDIA APPEARANCES

National Radio Astronomy Observatory NSF NRAO Announces 2026 Jansky Fellows	2026
Green Bank Observatory Students Contribute to New Understanding of “Twinkling” Pulsars	2025
West Virginia University WVU faculty, students contribute to cosmic breakthrough uncovering evidence of low-frequency gravitational waves	2023

PANELS

Walter Payton College Preparatory High School <i>Alumni STEM Panel</i>	March 2024
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AWARDED GRANTS

West Virginia University Eberly College of Arts & Sciences Travel Grant <i>3V459 A. Keith and Sandra F. McChung Enrichment Endowment</i> Principal Investigator	2022 \$600
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HONORS AND AWARDS

National Radio Astronomy Observatory <i>Jansky Postdoctoral Fellowship</i>	2026-Present
Green Bank Observatory <i>Green Bank Observatory Postdoctoral Fellowship</i>	2023-2026
Oberlin College <i>Oberlin College Department of Physics & Astronomy Honors Program</i>	2016-2017
Oberlin College <i>John Frederick Oberlin Scholarship</i>	2013-2017

ORGANIZATIONS

- North American Nanohertz Observatory for Gravitational Waves (NANOGrav): *Full Member*
- International Pulsar Timing Array (IPTA): *Full Member*
- American Astronomical Society: *Full Member*

SKILLS

- **Programming Languages:** Python, Bash, C shell, Unix/Linux, HTML
 - **Scientific Python Packages:** Numpy, Scipy, Matplotlib, Astropy, PyCyc, Scintools, Pypulse
- **Software Packages:** Simulink, L^AT_EX, TEMPO/TEMPO2, PSRCHIVE, DSPSR, Slurm, Jupyter/IPython